

SMILES strings

Ontology classes

<chem>CC(CC/C=C\C=C=C\1/O[C@H]1)C</chem>				✓	
<chem>Cl.COc1ccc(OC)c(c1)C(O)C</chem>	✓		✓		✓
<chem>CC(=O)Oc1ccc(\C=C\1C=O)c</chem>		✓		✓	
<chem>O1C2(C1CCC(C3C(C(C3)(C)</chem>				✓	

Inputs

Labels

ELECTRA

Boxes

Points

A

B

C

D

E

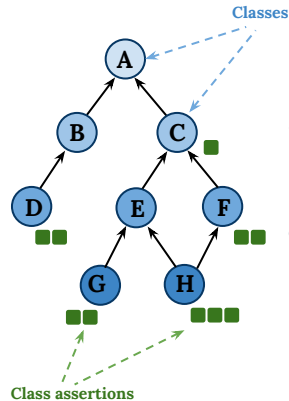
F

G

H

Embedding space
Geometric regions

Backpropagation of point-in-box errors



Specific to general

Extended ontology
Symbolic semantics

Training data
implicit label dependencies

AI Model
learn label dependencies